ECE 3317
Applied Electromagnetic Waves

Prof. David R. Jackson

Fall 2026

Notes 1
Introduction

Adapted from notes by Prof. Stuart A. Long

Why study electromagnetic waves?

Motivation:

Most basic of all ECE courses: *Electromagnetics provides the foundation for all electrical sciences and engineering. All of circuit theory and power engineering is a special case (low frequency, where dimensions are small relative to a wavelength).*

Electromagnetics explains physical phenomena: *What is light? What are radio waves? How do electric and magnetic fields behave?*

Important to know about EM: *It is extremely important for areas such as wireless communications, microwave engineering, and RF design. It is also very important for power engineering and micro/nano-electronics. It is good to know for all areas of ECE.*

Applications

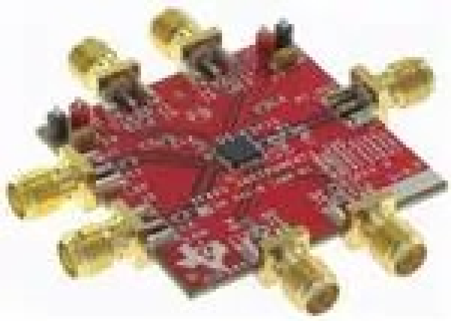
Antennas

Radar

Wireless Communications

RFID

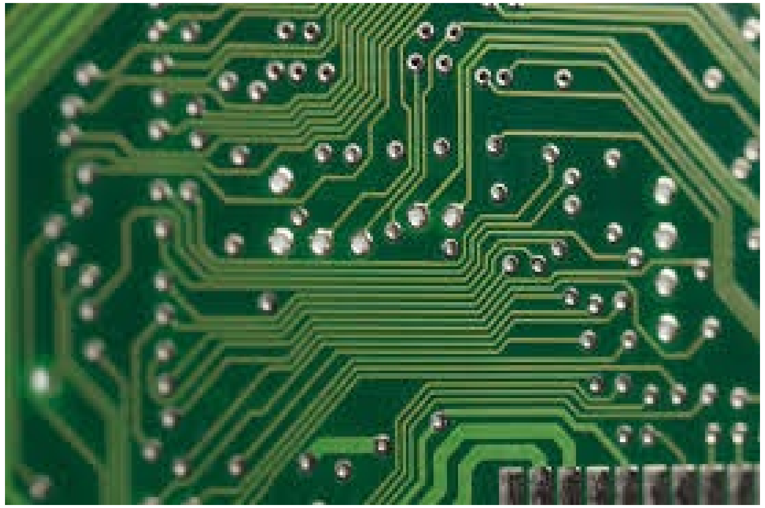
RF circuits

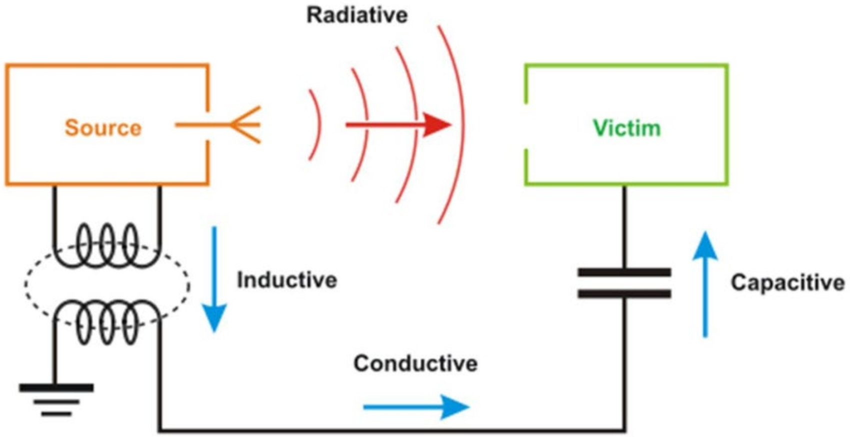


Applications (cont.)

***Computer and Electronic Applications*:**

- *At higher frequencies <u>transmission line effects</u> become more important. It becomes necessary to model the electromagnetic performance of the system (simple circuit theory is no longer adequate).*
- *Electromagnetic Compatibility (EMC) and Electromagnetic Interference (EMI) also become important at high frequency due to <u>radiation</u> and <u>coupling effects</u>.*





Course Theme

- In ECE 3317, we will deal with time-varying fields.

- In ECE 3318, you become familiar with static fields. (The methods learned there are accurate for power frequencies.).

More specialized courses (these require ECE 3317):

- **ECE 5317: Microwave Engineering**
  Prereq: ECE 3317. Transmission lines, waveguides, microstrip circuits, microwave circuit theory, scattering matrices, resonators, impedance transformers, and filters.

- **ECE 5318: Antenna Engineering**
  Prereq: ECE 3317. Antenna concepts, linear wire antennas, linear arrays, aperture and horn antennas, microstrip antennas, dielectric resonator antennas, frequency-independent antennas, and measurement techniques.

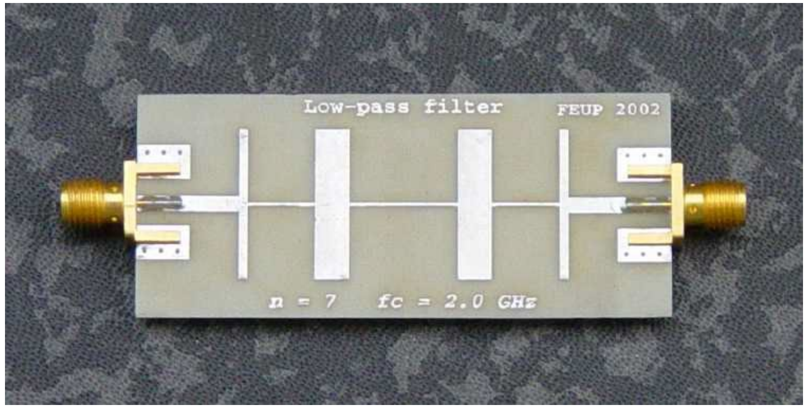
Course Theme (cont.)

ECE 3317, ECE 5317, ECE 5318

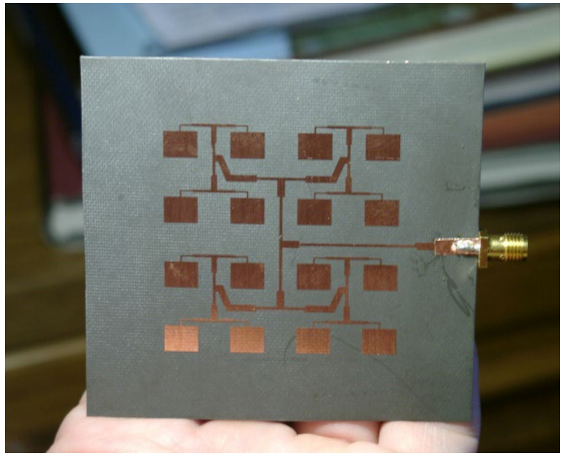
A microwave integrated circuit.

High Frequency

A cell-phone base-station antenna.



A microwave filter constructed from microstrip.



A microstrip antenna array.

Course Theme (cont.)

ECE 3318

Power buses in a substation.

A transformer in a substation.

Low Frequency

Overhead high-voltage power lines

Large AC generators at Hoover Dam.

Homework

- Homework is normally assigned once per week. It is due at the <u>beginning</u> of class on the due date (submitted in class).
- Homework will be distributed via the class Canvas site.
- Please staple it, fold it (vertically), and write on the outside the following: the class number (ECE 3317), the name of instructor (Prof. Jackson), the homework assignment number (e.g., HW 1), and your name.
- No late homework is accepted.

Announcements

- All important announcements will be placed on the class Canvas site in the "Announcements" section. It is your responsibility to check this often.
- Sometimes emails will be sent to the class, so it is your responsibility to make sure that your university email is working properly (and forwarded properly, if appropriate).

Announcements (cont.)

- Please read the syllabus carefully! You are responsible for everything on it! (It is on the Canvas site.)

- Please read the UH Academic Honesty Policy in the online version of the UH Student Handbook (the website is in the syllabus).

- Fill out the Academic Honesty and Syllabus Form, sign it, and return it by the deadline indicated (Sep. 9, 2026) or else you may be dropped. The form is on the Canvas site.

Announcements (cont.)

Class attendance is required

Class attendance will be taken.

- If you have three <u>unexcused</u> absences, your grade at the end of the semester may be lowered.

- To have an absence excused, you must get permission from the instructor <u>before</u> class begins.

Class Notes

The class notes are on the class Canvas website (both pptx and pdf versions).

If you have trouble reading the equations in the pptx version, you are most likely having a font problem due to missing MathType fonts or the Handscript SF font (used to represent time-domain vector fields).

Here is what you can do:

- Use the pdf version of the class notes.
- Install MathType (see next slide).
- Download the Handscript SF font (placed on Canvas site under "Handouts").

MathType

❖ In order to obtain a free license, please fill out and submit the form that is found at the link below:

https://forms.office.com/r/sd0agNuawV

MathType License Request

MathType is the professional version of the Equation Editor in Microsoft Word. It allows you to create complex equations with a simple point and click. MathType works with any word processor, a presentation program (such as Powerpoint), Canvas, page layout programs, HTML-authorizing tools, and other software, to create equations for research papers, class material, web pages, slide presentations, dissertations and journal articles.

- May only be used by current UH faculty and students.
- May be used on UH owned computers or devices owned by UH faculty and students.
- It is exclusively for academic use.
- May not be used for commercial gain.
- May only be used on one device at a time.
- Must be uninstalled when you are no longer affiliated with the University.

For more information about MathType, please see:
https://www.egr.uh.edu/sites/ccoe.egr.uh.edu/files/files/mathtype_for_making_and_managing_equations.pdf

For more information please contact Arturo Padilla, apadilla@uh.edu.

MathType

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https://www.egr.uh.edu/sites/ccoe.egr.uh.edu/files/files/mathtype_for_making_and_managing_equations.pdf

MathType for Making and Managing Equations

Introduction to MathType

MathType is software tool that allows you to easily create and manage equations inside of a Word document or a PowerPoint document. It is perfect for making equations inside of a thesis or dissertation, assuming that you will be using Microsoft Word to write your thesis or dissertation (If you use Latex to write your thesis or dissertation, then you do not need MathType.) It is also great for making equations inside of a PowerPoint presentation. When you install MathType, it will automatically create a tab within Microsoft Word and Microsoft PowerPoint, so that you can then easily use MathType when you are working on a Word or PowerPoint document.

Reading Assignments

Reading assignments are posted on the Canvas site from the required text (Hayt & Buck book).

- Please follow along with the class notes and read the corresponding sections in the book.
- Ideally, it would be good to read the appropriate sections in the book both before and after seeing the lecture.